

1. Explain what a "database" is in simple terms. What are some of the advantages of using a database like PostgreSQL for a web application.

a. A database is a platform or system for storing, organizing, and managing data so that it can be easily retrieved and updated when needed.

b. These are some of the advantages of using PostgreSQL for a web application:

- Reliability: PostgreSQL is highly reliable and can handle large data without crashing.
- PostgreSQL makes it easier to perform **complex data operations** efficiently.
- It ensures that the stored data is **protected**, with features like encryption and user authentication.
- Helps multiple users **read and write data simultaneously** without conflicts or slowdowns.

2a. What is docker

Docker is a tool that allows developers to package applications and their dependencies into containers. It is used for CI/CD mode of application deployment.

2b. Why is it useful for developers

- It helps applications to run identically on all platforms and environments. Whether on Windows, Mac, or Linux, Docker ensures that applications work the same way everywhere.
- Makes collaboration with other developers easy and makes deployment easy.
- Many modern apps use microservices architecture (breaking an app into small services). Docker makes it easy to manage and scale these services independently.
- Continuous Integration/Continuous Deployment (CI/CD) benefits from Docker by ensuring that code is tested and deployed consistently.

2c. Have you ever used Docker or seen it used in a project

Yes I have worked with docker

3. Imagine you're building a simple web application to track your favorite movies. What kind of data would you need to store, and how might you organize it in a database?

If I'm building a simple web application to track favorite movies, I'll need to store different types of data related to movie details, users, and their interactions. This is how I might organize it in a relational database like PostgreSQL

Movies Table

Fields : id (Primary Key, Unique Movie ID), title (Movie Title), release_year (Year of Release), genre (Genre, e.g., Action, Comedy), director (Director's Name), rating (User Rating), description (Short description), created_at (Timestamp).

User Table

Fields: id (Primary Key, Unique User ID), username (Unique Username), email (User's Email), password_hash (Encrypted Password), created_at (Timestamp)

Favorites Table

Fields: id (Primary Key), userId (Foreign Key ref. Users Table), movieId (Foreign Key ref. Movies Table), added_at (Timestamp)

4. Tell us about a personal project you've worked on using Python. What did you learn from the experience?

One personal project I worked on recently using Python was building a **Sales and Inventory Management System** using Django and PostgreSQL database. The system was built to implement point of sale, manage inventory, manage customers, generate sales reports, and reveal certain statistical summaries of sales reports using charts on admin sales dashboard.

Technologies and Programming languages used are Python, Django Framework, External API, Javascript, DataTable, Chart.js PostgreSQL, Git and Github, Django templates, html, css, bootstrap, jquery.

What I Learnt

- I learnt how to model a more robust database system for a large application such as this. I also learned how to simplify more complex database structures.
- Make proper dashboard, displaying monthly profit/loss, total products sold, top 5 products by sales, total number of customers and some other statistics using charts.

- Optimize performance. The first time I deployed it, it was slow to load the pages. But I figured out why and fixed the problem and the system runs faster without any effect on the functionality.
- More importantly, I learnt how to approach problems iteratively, test different algorithms, and optimize performance.

5. Please list the three things you dislike most about each of PostgreSQL and Python, and explain why you dislike those things.

PostgreSQL

- Configuration is a bit complex for beginners. Dependency of Psycopg2 on other libraries makes its installation difficult at times.
- setting up high availability and horizontal scaling is more difficult in PostgreSQL compared to databases like MongoDB.
- Indexing and ranking results efficiently in PostgreSQL often require additional tweaking

Python

- Slower execution compared to other high level languages like Java which are compiled, because Python is interpreted.
- Performance can be an issue for computationally heavy tasks, requiring optimization techniques like Cython or NumPy vectorization because of its low speed.
- Python has version compatibility issues also, one of the things I don't like about it

